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Volatility forecasting and assessing risk of financial markets using multi-transformer neural network based architecture

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A B S T R A C T

This research introduces a more reliable model for predicting market volatility. The model incorporates Transformer and Multi-transformer layers with the GARCH and LSTM models and compares their performance to classic GARCH-type models. Euro-US Dollar FX Spot Rate (EURO-USD), Australian-US Dollar FX Spot Rate (AUD-USD), S&P 500 Index, FTSE 100 Index, Reliance Industries Ltd., and Samsung Electronics Co Ltd. are analyzed. The timeframe examined is January 2005 to December 2021, with training from 2005 to 2016 and testing from 2017 to 2021. Empirical evidence suggests that hybrid Neural-network models, notably Transformer-based models, outperform individual transformers, deep learning, neural networks, and traditional GARCH-type models, even in unpredictable conditions like the COVID-19 pandemic. Within the hybrid Neural-network models, MT-GARCH and MTL-GARCH showed lower validation error (RMSE) than the other models, showing that the bagging mechanism added to the attention mechanism of the Multi-Transformer architecture helped to lower the error in the variance in the noisy data of the daily returns of the assets, reducing the RMSE of the hybrid multi-Transformer models. In addition, three to four of the five hybrid neural-network models showed appropriate risk estimates for the entire test period, as observed from the Christoffersen test. Moreover, the lengthy sample period helps test whether hybrid models perform better than classic GARCH-type models in volatile conditions like the COVID-19 pandemic.

1. Introduction

The impact of the Global Financial Crisis of 2007-08 in terms of the risk management frameworks for financial institutions around the world has changed significantly, which include higher capital and funding requirements, controlled leverage, higher liquidity and higher risk reporting standards like the BCBS 239 (Härle et al., 2015). Despite these risk management frameworks which have been put in place, financial institutions face significant losses due to unexpected events like the COVID-19 pandemic. Therefore, using market risk and equity risk models that make accurate forecasts has become essential. Among the parameters that affect equity risk, stock volatility forecasts are one of the key parameters. According to the Markowitz (1952) model, this volatility can be measured as the standard deviation of the returns of the asset. However, the heteroskedasticity in financial time-series makes it difficult to forecast the conditional variance accurately (Engle, 1982).

Traditionally, GARCH (Generalized Auto-Regressive Conditional Heteroskedasticity)-type models (Engle, 1982; Bollerslev, 1986; Gokcan, 2000; Alberg, Shalit, & Yosef, 2015) like GARCH (1,1), Exponential GARCH, GJR-GARCH and Threshold GARCH have been used to forecast volatility which accounted for volatility clustering effects (Mandelbrot, 1963). The results from such models were quite successful in predicting one period forecast variance using the information from the recent past. A major drawback of these GARCH models was that they ruled out any negative correlation between current returns and future returns volatility. Further, the GARCH models impose parameter restrictions which could limit the dynamics of the conditional variance process. Nelson (1991) solved these drawbacks by introducing the exponential GARCH model. Other models like GJR-GARCH (Glosten et al., 1993) exploited the seasonal pattern in volatility.

Another drawback of these purely econometric models is that they are based on the assumption that the underlying financial time-series is stationary, which may not always be the case. For this reason, machine learning and deep learning models gained relevance. Hamid and Iqbal (2004) used artificial neural networks (ANNs) to forecast volatility of the S&P 500 index future prices, and the results outperformed the

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implied volatility forecasts. Gonzalez Miranda and Burgess (1997) also used ANNs to forecast volatility, and found the results to dominate the traditional econometric methods due to the ability of neural networks to understand the non-linear relationships of the variables.

Many other recent studies (Rezaei et al., 2021; Ghosh et al., 2021) use machine learning and deep learning methods to forecast volatility since these models are able to learn and manipulate the model parameters even for data inputs that are incomplete or fuzzy. Another category of neural networks called recurrent neural networks (RNNs) have gained relevance, because of their ability to learn temporal patterns in the data due to the presence of a feedback loop. The use of neural networks on financial time-series data isn't uncommon and has been used to predict market trends by combining approaches from both technical and fundamental schools of thought (Picasso, et al., 2019). The Long short-term Memory model (LSTM) is one such RNN model which has outperformed both GARCH type models and traditional machine learning models (Yu and Li, 2018; Fischer and Krauss, 2018). Malandri et al. (2018) utilized public financial sentiment along with LSTM model for portfolio selection, thus highlighting the possible uses of LSTMs in other areas such as portfolio management. As shown by literature, neural networks have shown to predict volatility more accurately, inspiring us to use neural networks in this paper.

There have been growing instances of hybrid models being used to forecast volatility with encouraging results. Roh (2007) combined ANNs with GARCH, EGARCH and EWMA to predict the volatility of the Korean stock index. In that study the combined models NN-GARCH and NN-EGARCH had a lower mean absolute error than the solitary neural network model. In many recent studies, hybrid models combining GARCH models with LSTMs and ANNs (Kristjanpoller et al., 2014; Lu et al., 2016; Bhattacharya and Ahmed, 2018; Kim and Won, 2018; Hu et al., 2020; Kakade, Jain and Mishra, 2022; Kakade et al., 2022a,b) have also been used to predict stock volatility which are found to further improve the accuracy of the volatility forecasts compared to the individual LSTM and ANN models. Liu, Oosterlee & Bohte (2019) used data-driven approaches by the means of ANNs to value options and calculate implied volatilities, finding the ANN models reduce the computation time significantly. Liang et al. (2020) used data-driven exponentially weighted moving averages (DD-EWMA) for volatility forecasts and stock price forecasts for algorithmic trading.

Forecasting stock market volatility has been of interest not just to financial organizations and regulators, but also to academics. As financial markets are susceptible to sharp, unexpected declines, it is very desirable to use models that accurately predict volatility. Additionally, it is advantageous to have indicators that appropriately quantify risk.

Very recently, there have been mentioning of literature relating to Transformer and Multi-Transformer models. Multi-Transformer gathers diverse subsets of training data at random and combines several multi-head attention processes to generate the result. Following the logic of bagging, the purpose of this design is to increase the attention mechanism's stability and precision. What differentiates these transformer models from the traditional models is their self-attention mechanism which helps in showing improved accuracy compared to the traditional models like LSTM, which is also shown by Devlin et al. (2018) for language processing tasks. Ramos-Pérez, et al. (2021) combined transformer models with both GARCH models and LSTM models in order to form different hybrids and compare the performance in forecasting the volatility of S&P 500. They found that the Multi-Transformer and GARCH hybrid model significantly outperformed the solitary GARCH model's performance, and also gave more accurate results than the LSTM-GARCH models. Transformer layers are the state of the art in natural language processing. Xing, Zhang & Cambria (2019) used a Sentiment-aware volatility forecasting method, which incorporates market sentiment along with traditional statistical and AI approaches, and found it to outperform both GARCH-type models and neural networks.

This study extends the work of Ramos-Pérez, et al. (2021) that provides the architecture of four hybrid models namely; (i)Transformer-GARCH model (T-GARCH); (ii)Transformer-LSTM-GARCH model (TL-GARCH); (iii)Multi-Transformer-GARCH model (MT-GARCH) and (iv) Multi-Transformer-LSTM-GARCH model (MTL-GARCH) based on Transformer and Multi-Transformer layers. The results of the study provide more accurate stock market volatility models and appropriate equity risk measures by considering the case for six financial assets namely; Euro-US Dollar FX Spot Rate (Euro-USD), Australian Dollar-US Dollar FX Spot Rate (AUD-USD), S&P 500 Index, FTSE 100 Index, Reliance Industries Ltd., and Samsung Electronics Co. as compared to the benchmark deep-learning model like LSTM-GARCH model besides the traditional GARCH models. The timeframe examined is January 2005 to December 2021, with training from 2005 to 2016 and testing from 2017 to 2021. This study has a novel contribution in terms of showing improved forecasting of volatility using the Transformer-based models across various types of assets like foreign exchange rate, stock market indices and stock price data of individual companies. It is also found that even in extremely unpredictable situations like the COVID-19 pandemic, the hybrid Neural-network models, notably the Transformer-based models, outperform the individual classic GARCH-type models. This shows that the bagging mechanism added to the attention mechanism part of the Multi-Transformer architecture helped in solving problems like lowering the error in the variance of the noisy data characteristic of the daily returns in such assets. Further, this study goes beyond just testing the effectiveness of Transformer based models for various asset types in different volatile periods, but also evaluating it on different error metrics like Root Mean Squared Error(RMSE), Mean Absolute Error(MAE) and Expected Shortfall(ES). The lengthy sample period helps test whether hybrid models perform better than classic GARCH-type models in volatile conditions like the COVID-19 epidemic. Results also show that the risk assessments derived from these models are very accurate. The value at risk (VaR) of the majority of these models may be judged correct even in the year 2020, when the COVID-19 pandemic generated extraordinary market volatility. Christoffersen's (1998) Conditional Coverage test has been popularly used for backtesting VaR. Omari, C. , Mundia, S. and Ngina, I. (2020) recently used this test to backtest and forecast VaR for multiple financial assets in the extremely volatile COVID-19 period.

The rest of the paper is organized as follows: Section 2 discusses the dataset used in the study. Section 3 covers volatility forecasting methods based on Transformer and Multi-Transformer layers. Since this study introduces Multi-Transformer layers and adapts NLP Transformers for volatility forecasting, explanations on the theoretical underpinning of these structures are also presented. Section 4 provides the empirical results and offers a discussion of those results. Section 5 concludes.

2. Data and descriptive statistics

The time period considered for this study is from January 2005 to December 2021. The assets considered are Euro-USD FX Spot Rate, AUD-USD FX Spot Rate, S&P 500 Index, FTSE 100 Index, Reliance Industries Ltd. and Samsung Electronics Co Ltd. The prices considered for all the assets are of daily frequency. The issue of heteroskedasticity in the prices, which might arise in case the standard deviation is variable over time, is resolved by taking the natural logarithm of the closing prices. Further, the daily returns are computed as the first difference of the natural logarithm of the closing prices. Fig. 1 shows the time-series plot for the prices of all the six assets considered.

Table 1 shows the descriptive statistics for the prices of all the six assets considered. We observe that the mean of Euro-USD is higher than that of AUD-USD, while their standard deviation is the same. In the case of the S&P 500 index and the FTSE 100 index, both the mean and the standard deviation is found to be higher for FTSE 100 as compared to S&P 500. In the case of Reliance and Samsung, the latter is found to have both a higher mean as well as a higher standard deviation compared to Reliance. The positive skewness value for all the six assets shows that

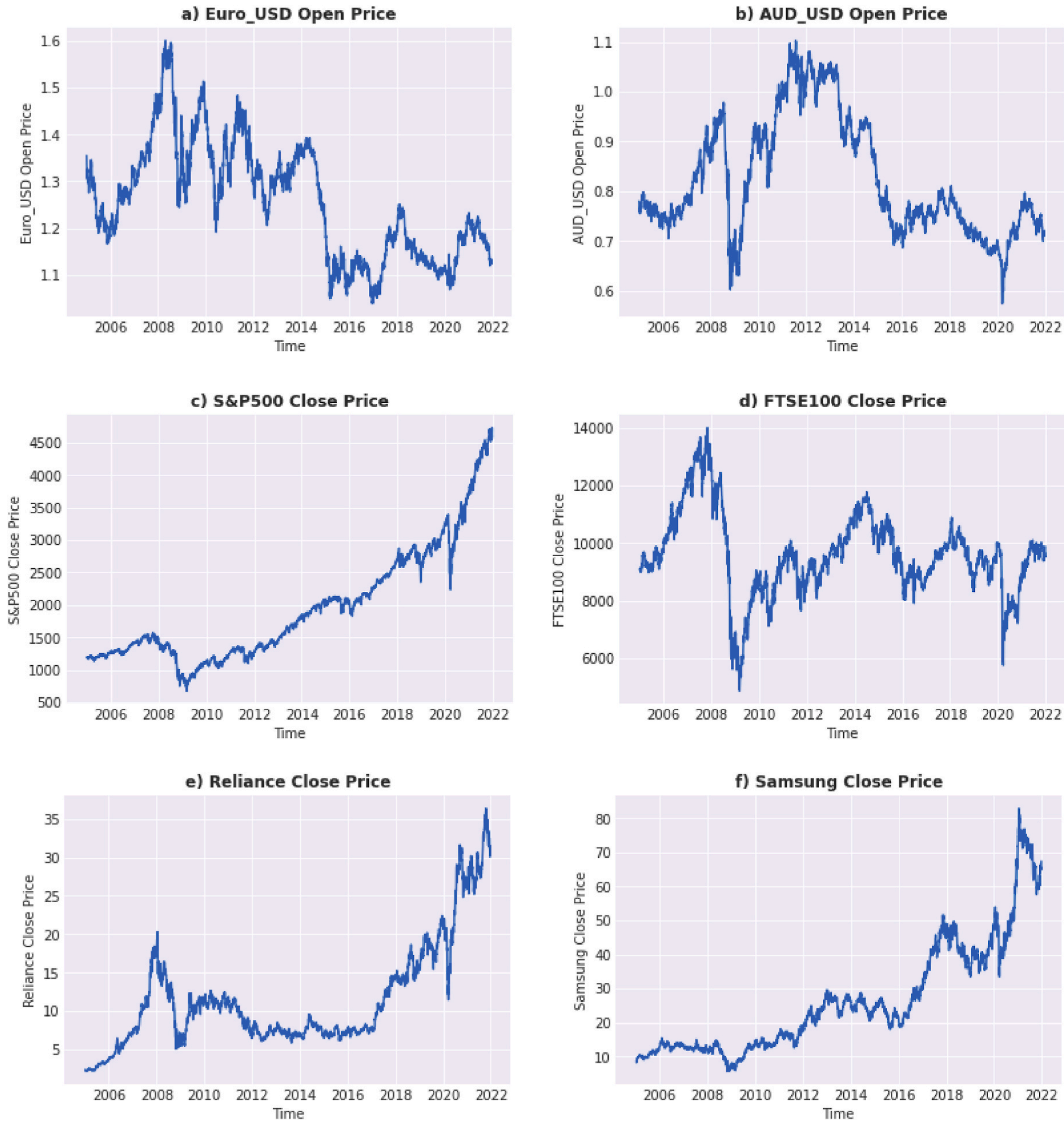

Fig. 1. Closing price of Euro-US dollar FX spot rate (Euro-USD), Australian dollar-US Dollar FX spot rate (AUD-USD), S&P 500 index, FTSE 100 index, Reliance Industries Ltd. and Samsung Electronics Co Ltd.

Table 1
Descriptive statistics.

Summary	Euro-USD	AUD-USD	S&P 500	FTSE 100	Reliance	Samsung
Mean	1.26	0.82	1954.74	9607.36	11.58	26.45
Std. Dev.	0.12	0.12	884.73	1440.84	6.95	16.46
Min.	1.04	0.57	676.53	4877.86	2.15	5.62
Max.	1.60	1.10	4725.79	13989.57	36.38	82.94
Skewness	0.38	0.61	1.12	0.13	1.39	1.17
Kurtosis	2.41	2.23	3.67	3.85	4.50	3.8

Note: Computed by authors.

they are skewed right i.e., the right tail is longer as compared to the left tail. In addition, the kurtosis value for Euro-USD and AUD-USD indicates that the tails are slightly thinner, i.e., more values are located around the mean rather than the tails of the distribution while the kurtosis value for S&P 500, FTSE 100, Reliance and Samsung indicates that tails are thicker i.e., more values are located in the tails of the distribution as compared to around the mean.

3. Methodology

The time period considered for the study is from January 2005 to December 2021. A rolling window methodology has been used to fit the models on our dataset, and then the optimum model configuration is chosen by looking at the smallest error (RMSE). The training set is taken from January 2005 to December 2016 to fit the models and the remaining dataset forms the test set.

Rolling window uses a fixed block of observations to fit the data, and then used to calculate the forecast for the next step. Hence, the model is applied repeatedly on subsets of the full dataset. The rolling window size chosen is 500 trading day. Hence the proposed and benchmark models are fitted using the last 500 data points and using that the volatility of the next 1-day is calculated.

The input variables used are daily log returns and the standard deviation of the last five daily log returns, thus taking n as 5:

$$\sigma_{t-1} = \sqrt{\frac{\sum_{i=1}^{n} (r_{t-i} - E[r])^2}{n-1}} \tag{1}$$

For LSTM, transformers and Multi-Transformer layers, a lag of the last 10 observations is considered as input to fit the layers.

For validating the forecasts made by the different models, the mean absolute value (MAE) and the root mean squared error ($RMSE$) have been used:

$$MAE = \sum_{t=1}^{N} \frac{\left|\sigma_{i,t} - \widehat{\sigma}_{i,t}\right|}{N} \tag{2A}$$

$$RMSE = \sum_{t=1}^{N} \frac{\left(\sigma_{i,t} - \widehat{\sigma}_{i,t}\right)^2}{N} \tag{2B}$$

where N is the total number of observations.

The benchmark model used is the GARCH model. For this we assume that the innovations (ϵ_t) would follow a t-distribution. The model is parametrized by the values of p and q, and hence is called the GARCH (p, q) model as shown below:

$$\hat{\sigma}_t^2 = \omega + \sum\nolimits_{i=1}^{q} \alpha_i r_{t-i}^2 + \sum\nolimits_{i=1}^{p} \frac{\beta_i \sigma_{t-i}^2}{\hat{r}_t} = \hat{\sigma}_t \epsilon_t \tag{3}$$

here, σ_{t-i}^2 is the last observed volatility, r_{t-i} is the previous returns, and ω, α, and β are the parameters to be estimated. The returns generated by this model would also follow a conditional t-distribution.

Another benchmark model is considered. These are the hybrid models based on merging the GARCH model and LSTMs. LSTMs are known to be able to handle time series better, and hence we take a lag of last ten observations as input for the LSTM-GARCH model, both for the returns and standard deviations.

To train the models', true implied volatility is used. True implied volatility is quantified by taking the standard deviation of the future log returns as follows:

$$\hat{\sigma}_{i,t} = \sqrt{\frac{\sum_{n=0}^{i-1} \left(r_{t+n} - E\left[r_f\right]\right)^2}{i-1}} \tag{4}$$

where $E[r_f] = \sum_{n=0}^{i-1} r_{t+n}/i$ and $i = 5$.

For LSTM-GARCH, the input is processed by three layers. First is the 32-unit LSTM layer, followed by two feedforward layers having 8 and 1 neurons, respectively (see Fig. 2).

Next, we look at the transformer architecture and how we can apply it in the context of time series. The transformer uses an encoder-decoder architecture. The transformer's encoder extracts features from input data, while the decoder utilizes these features to generate output data (translation). Comprising multiple blocks, the encoder-decoder mechanism processes input through encoder blocks, with the last block's output serving as input features for the decoder.

One key component of transformer models is the use of positional encoders. The input fed to the transformer models is encoded so as to contain information about its relative position within the time series. This step is crucial in transformer models because, unlike LSTMs, they do not have a recurrence structure.

Here, we use a positional encoder, which only depends on the lag of the input variables. Transformers use a smart positional encoding scheme, where each position/index is mapped to a vector. Hence, the output of the positional encoding layer is a matrix, where each row of the matrix represents an encoded object of the sequence summed with its positional information.

As is the general norm, a wave function is used as the positional encoder:

$$PE_{pos} = cos\left(\pi \frac{pos}{N_{pos}-1}\right) = sin\left(\frac{\pi}{2} + \pi \frac{pos}{N_{pos}-1}\right) \tag{5}$$

here, we take pos as the position of the observation in the time series and N_{pos} as the maximum possible lag.

The other key aspect of transformer architecture is multi-head attention. Multi-Head attention divides the variables into separate 'heads' in order to run the different scaled dot-product attention units in parallel. A scaled dot-product attention unit is defined as follows:

$$Attention\ (Q, K, V) = \left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{6}$$

$$MultiHead\ (Q, K, V) = Concat\ (\ head\ _1, \ldots,\ head\ _h) W^O \tag{7}$$

$$head\ _i = Attention\ \left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{8}$$

After calculating the different heads, they are concatenated and fed to the next stage which includes a normalization layer, a feedforward layer with the ReLU activation function and another normalization layer.

The T-GARCH hybrid model suggested here, takes the inputs of the GARCH model and feeds it into the transformer layer. Similarly, a TL-GARCH model takes the inputs of the LSTM-GARCH model and feeds it into the transformer layers (see Fig. 3). For LSTM-GARCH, there is no need of positional encoders as the LSTM layers are able to recognize the structure of the data.

The initial learning rate is set to 0.01. Then, we use the Adaptive Moment estimator (ADAM) to gradually update the weights of the different layers, by progressively changing the initial learning rate. We optimize the level of dropout regularization using mean squared error loss function. The same is used for backpropagation as well. The batch size is fixed at 64 and we use 100 epochs to get the final outputs.

In the Multi-Transformer architecture suggested we randomly select 90% of the dataset, to generate T different random samples, each having one multi head attention unit. The positional encoder used remains the

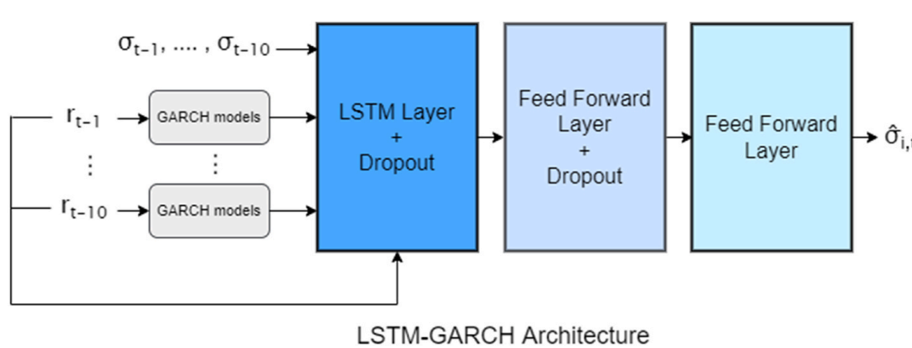

Fig. 2. LSTM-GARCH Architecture. *Notes:* This figure shows a flowchart of the architecture of the hybrid neural network model LSTM-GARCH.

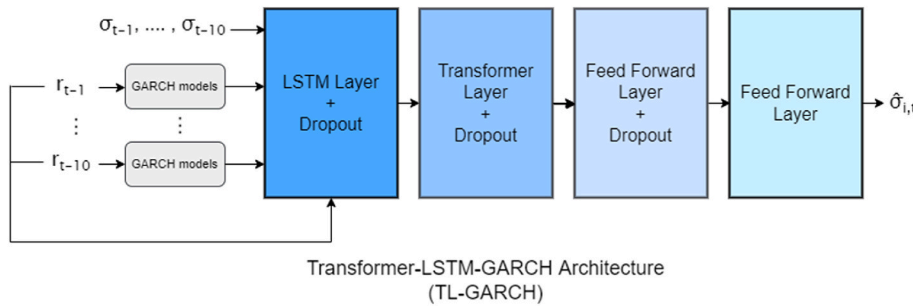

Fig. 3. Transformer-LSTM-GARCH Architecture. *Notes:* This figure shows a flowchart of the architecture of the hybrid neural network model Transformer-LSTM-GARCH (TL-GARCH).

same as was in transformer model. The final attention matrix is calculated by taking the average of all the units. This is known as the Average Multi-Head mechanism:

$$Average\ MultiHead(Q,K,V) = \frac{\sum_{t=1}^{T} Concat\left(head_{1,t},\ldots,head_{h,t}\right) W_t^O}{T} \tag{9}$$

$$head_{i,t} = Attention\left(Q_t W_{i,t}^Q, K_t W_{i,t}^K, V_t W_{i,t}^V\right) \tag{10}$$

Next, bagging is applied to the attention mechanism. Bagging is a common technique applied in neural networks to reduce variance in a noisy dataset. It involves creating multiple datasets through random sampling with replacement from the training set. Our models are then independently trained on these samples, and the final estimate is derived from the aggregated predictions, improving accuracy. Since bagging is computationally very expensive, we do not apply it to all the input data, and instead just apply it to the attention mechanism. Bagging helps in significantly dropping the error variance and reducing over-fitting thus leading to higher accuracy.

Finally, we look at Multi-Transformer hybrid models. The MT-GARCH feeds the output of the GARCH model into the Multi-Transformer layers. Similarly, MTL-GARCH feeds the output of the LSTM-GARCH into the Multi-Transformer layers (see Fig. 4).

4. Results and discussions

4.1. Analysis of the results for the training period based on hybrid neural -network models

The period considered for training the hybrid Neural-network based models is from January 2005 to December 2016. As highlighted in Section 3, the approach of rolling windows is used to fit each model. The hyperparameters considered are a confidence level of 95%, 100 epochs and a learning rate of 1%. For each model, the parameter being optimized for is the dropout ratio. Table 2 shows the RMSE of each model for the dropout ratios of 0%, 5%, 10% and 15% respectively. The dropout ratio provides regularization, which prevents the overfitting of the models. The process of optimization shows that a dropout of 0% shows higher RMSE values compared to the other dropout ratios, irrespective of the type of the model. This indicates that models that are based on architectures like LSTMs and Transformers require an appropriate level of regularization, so as to avoid overfitting. The optimal dropout level obtained is the one, which shows the lowest RMSE for each model and dataset. Tables 2–7 show the RMSE according to the model and dropout ratio for all the six assets considered. Table 8 shows the optimal dropout ratios obtained for all the models and the six datasets.

Table 2
RMSE by model and dropout ratio for the training set of Euro-USD.

Model	Dropout = 0%	Dropout = 5%	Dropout = 10%	Dropout = 15%
LSTM-GARCH	2.5260E-04	1.5243E-04	1.1601E-04	9.4182E-05
T-GARCH	1.2159E-04	1.2942E-04	4.2471E-05	4.5754E-05
TL-GARCH	9.5105E-05	6.2898E-05	4.6792E-05	3.6994E-05
MT-GARCH	3.5424E-04	2.0308E-04	1.2268E-04	1.3832E-04
MTL-GARCH	9.8896E-05	5.5725E-05	3.8298E-05	3.4214E-05

Notes:
a This table shows the RMSE (Root Mean Squared Error) for all the hybrid Neural-network models for the training set of the financial asset for the dropout ratios of 0%, 5%, 10% and 15% respectively. The dropout ratio which shows the lowest RMSE is chosen for that particular model.
b The names used against the models in acronym forms are (i) The Long short-term Memory-Generalized Auto Regressive Conditional Heteroskedasticity model (LSTM-GARCH); (ii)Transformer-GARCH model (T-GARCH); (iii)Transformer-LSTM-GARCH model (TL-GARCH); (iv)Multi-Transformer-GARCH model (MT-GARCH) and (v) Multi-Transformer-LSTM-GARCH model (MTL-GARCH).

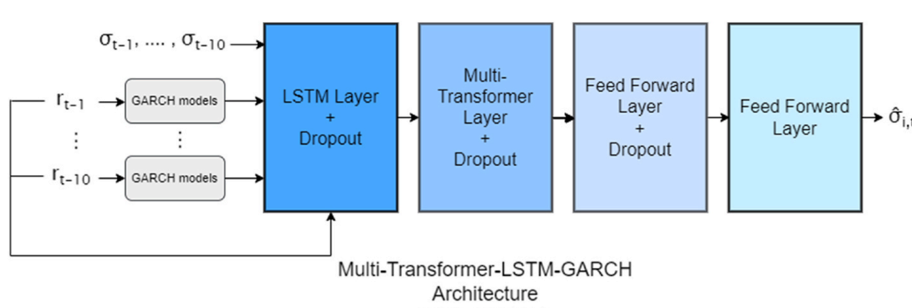

Fig. 4. Multi-Transformer-LSTM-GARCH Architecture. *Notes:* This figure shows a flowchart of the architecture of the hybrid neural network model Multi-Transformer-LSTM-GARCH (MTL-GARCH).

Table 3
RMSE by model and dropout ratio for the training set of AUD-USD.

Model	Dropout = 0%	Dropout = 5%	Dropout = 10%	Dropout = 15%
LSTM-GARCH	3.1986E-04	2.1573E-04	1.7534E-04	1.3473E-04
T-GARCH	1.0747E-04	7.6971E-05	4.8283E-05	4.6017E-05
TL-GARCH	9.7433E-05	6.1910E-05	5.6861E-05	4.7936E-05
MT-GARCH	1.4693E-04	1.5447E-04	7.7297E-05	4.8365E-05
MTL-GARCH	9.4246E-05	6.1895E-05	5.1085E-05	5.1372E-05

Notes: Same as Table 2.

Table 4
RMSE by model and dropout ratio for the training set of S&P 500.

Model	Dropout = 0%	Dropout = 5%	Dropout = 10%	Dropout = 15%
LSTM-GARCH	12.2748E-05	8.23825E-05	7.44303E-05	7.85252E-05
T-GARCH	12.202E-05	8.10432E-05	7.38805E-05	7.60479E-05
TL-GARCH	10.9796E-05	8.47248E-05	7.81988E-05	8.22729E-05
MT-GARCH	11.1646E-05	8.33523E-05	7.26385E-05	7.95651E-05
MTL-GARCH	10.6129E-05	8.61647E-05	7.74765E-05	8.6369E-05

Notes: Same as Table 2.

Table 5
RMSE by model and dropout ratio for the training set of FTSE 100.

Model	Dropout = 0%	Dropout = 5%	Dropout = 10%	Dropout = 15%
LSTM-GARCH	12.0557E-05	9.0821E-05	8.85062E-05	8.71266E-05
T-GARCH	12.3884E-05	10.6654E-05	8.96913E-05	9.44749E-05
TL-GARCH	10.6356E-05	8.54304E-05	9.47775E-05	10.6872E-05
MT-GARCH	11.1748E-05	8.87079E-05	8.35294E-05	10.1722E-05
MTL-GARCH	10.8733E-05	8.60283E-05	8.82084E-05	10.297E-05

Notes: Same as Table 2.

Table 6
RMSE by model and dropout ratio for the training set of Reliance.

Model	Dropout = 0%	Dropout = 5%	Dropout = 10%	Dropout = 15%
LSTM-GARCH	1.9441E-04	1.6144E-04	1.7832E-04	1.9598E-04
T-GARCH	1.9377E-04	1.7973E-04	1.7680E-04	2.1451E-04
TL-GARCH	1.5662E-04	1.5506E-04	1.8189E-04	2.1862E-04
MT-GARCH	1.7147E-04	1.4930E-04	1.6940E-04	2.1512E-04
MTL-GARCH	1.5507E-04	1.5426E-04	1.8364E-04	2.1857E-04

Notes: Same as Table 2.

Table 7
RMSE by model and dropout ratio for the training set of Samsung.

Model	Dropout = 0%	Dropout = 5%	Dropout = 10%	Dropout = 15%
LSTM-GARCH	1.7278E-04	1.3725E-04	1.4604E-04	1.7304E-04
T-GARCH	1.4973E-04	1.3767E-04	1.4874E-04	1.8603E-04
TL-GARCH	1.3751E-04	1.3702E-04	1.5522E-04	2.0182E-04
MT-GARCH	1.5590E-04	1.3960E-04	1.7544E-04	2.1502E-04
MTL-GARCH	1.3520E-04	1.2633E-04	1.5725E-04	1.9818E-04

Notes: Same as Table 2.

Table 8
Optimal dropout ratios for each model and asset.

Model	Euro-USD	AUD-USD	S&P 500	FTSE 100	Reliance	Samsung
LSTM-GARCH	15%	15%	10%	15%	5%	5%
T-GARCH	10%	15%	10%	10%	10%	5%
TL-GARCH	15%	15%	10%	5%	5%	5%
MT-GARCH	10%	15%	10%	10%	5%	5%
MTL-GARCH	15%	10%	10%	5%	5%	5%

4.2. Analysis of the results for the testing period based on hybrid neural -network models

The period considered for testing the hybrid Neural-network based models is from January 2017 to December 2021. Tables 9–14 show the RMSE of each model for each year and for the overall test period.

Out of all the hybrid Neural-network models, the Transformer-based models show the lowest RMSE for all the six assets considered for the entire test period. This is in line with Ramos-Pérez et al. (2021) who found that the Multi-Transformer architecture when combined with the GARCH model, performed better than the other hybrid models. This is because the mechanism of bagging (also known as bootstrap aggregation) reduced the variance in the noisy data of the daily returns of the assets, thereby decreasing the RMSE. Even though bagging is not generally applied to the Multi-Transformer architecture, applying it only to the attention mechanism part of the Multi-transformer architecture had the effect of lowering the RMSE. which can be seen in the

Table 9
RMSE by model and year for the test set of EURO-USD.

Model	2018	2019	2020	2021	2018–2021
GARCH	7.3425E-06	4.1950E-06	8.2334E-06	5.8437E-06	6.3167E-06
GJR-GARCH	7.3362E-06	4.2884E-06	7.8278E-06	6.0990E-06	6.2966E-06
TrGARCH	7.3543E-06	4.1604E-06	7.6535E-06	6.0901E-06	6.2133E-06
APGARCH	7.3442E-06	4.1884E-06	7.8736E-06	5.8787E-06	6.2242E-06
AVGARCH	7.2695E-06	4.0863E-06	7.8628E-06	5.6377E-06	6.1153E-06
FIGARCH	7.2981E-06	4.0508E-06	7.3886E-06	5.6691E-06	5.9867E-06
LSTM-GARCH	3.7333E-04	5.2788E-05	2.7976E-05	2.2105E-04	1.7622E-04
T-GARCH	1.7265E-04	2.4354E-05	4.2664E-05	5.4857E-05	7.7081E-05
TL-GARCH	2.0504E-05	1.3061E-05	2.8731E-05	3.9245E-05	3.2862E-05
MT-GARCH	2.3548E-04	1.4403E-04	4.2667E-05	4.8943E-05	1.8514E-04
MTL-GARCH	2.5140E-05	1.0161E-05	2.9010E-05	3.2257E-05	2.9506E-05

Notes:
a RMSE (Root Mean Squared Error) for all the hybrid Neural-network models as well as the individual traditional GARCH-type models, for each year in the test set and for the overall test period.
b The GARCH model is the GARCH (1,1) model; GJR-GARCH model is Glosten-Jagannathan-Runkle GARCH model; APGARCH model is the Asymmetric Power GARCH (1,1) model; TrGARCH model is the Threshold GARCH (1,1) model, FIGARCH model is fractionally integrated GARCH model and AVGARCH model is absolute value GARCH model.
c LSTM-GARCH model is The Long short-term Memory-Generalized Auto Regressive Conditional Heteroskedasticity model; T-GARCH model is the Transformer-GARCH model, TL-GARCH model is the Transformer-LSTM-GARCH model, MT-GARCH model is the Multi-Transformer-GARCH model and the MTL-GARCH model is the Multi-Transformer-LSTM-GARCH model.

Table 10
RMSE by model and year for the test set of AUD-USD.

Model	2018	2019	2020	2021	2018–2021
GARCH	9.6200E-06	5.1583E-06	1.7578E-05	7.4836E-06	1.0029E-05
GJR-GARCH	9.8326E-06	5.3708E-06	2.0382E-05	8.0862E-06	1.1068E-05
TrGARCH	1.0549E-05	4.8986E-06	1.6678E-05	7.8988E-06	9.9806E-06
APGARCH	1.0676E-05	5.3583E-06	2.2778E-05	8.3021E-06	1.1939E-05
AVGARCH	1.0468E-05	4.8551E-06	1.5318E-05	6.8544E-06	9.2965E-06
FIGARCH	1.0289E-05	5.2030E-06	1.6534E-05	7.8562E-06	9.9679E-06
LSTM-GARCH	4.0916E-04	5.5872E-04	3.0427E-05	2.1574E-04	2.9348E-04
T-GARCH	2.3808E-05	3.3169E-05	2.3994E-05	2.4078E-05	2.6531E-05
TL-GARCH	3.1423E-05	2.8278E-05	2.9082E-05	2.0001E-05	2.6847E-05
MT-GARCH	2.4973E-04	1.2173E-04	2.8774E-05	2.4649E-05	9.2504E-05
MTL-GARCH	2.4695E-05	3.6508E-05	2.7122E-05	3.4770E-05	3.1342E-05

Notes: Same as Table 9.

Table 11
RMSE by model and year for the test set of S&P 500.

Model	2018	2019	2020	2021	2018–2021
GARCH	8.2240E-05	5.1771E-05	1.0941E-02	3.6092E-05	3.1307E-03
GJR-GARCH	1.3525E-04	4.2298E-05	1.7775E-02	4.9619E-05	5.0722E-03
TrGARCH	1.3382E-04	3.4345E-05	1.5179E-02	4.8660E-05	4.3356E-03
APGARCH	1.6880E-04	5.0077E-05	1.8652E-02	5.5629E-05	5.3296E-03
AVGARCH	4.9299E-05	3.1143E-05	7.5444E-03	4.1226E-05	2.1607E-03
FIGARCH	7.4407E-05	4.9041E-05	1.0564E-02	3.5391E-05	3.0218E-03
LSTM-GARCH	6.3099E-05	5.4913E-05	2.2613E-04	6.1439E-05	1.0208E-04
T-GARCH	6.2608E-05	4.7417E-05	2.9674E-04	4.0749E-05	1.1858E-04
TL-GARCH	5.8428E-05	4.1272E-05	3.0366E-04	4.1736E-05	1.1839E-04
MT-GARCH	4.1893E-05	4.2796E-05	3.0826E-04	4.8969E-05	1.1937E-04
MTL-GARCH	1.0745E-04	4.2881E-05	2.9044E-04	6.7927E-05	1.3027E-04

Notes: Same as Table 9.

Table 12
RMSE by model and year for the test set of FTSE 100.

Model	2018	2019	2020	2021	2018–2021
GARCH	1.8068E-05	2.6703E-05	5.2647E-03	4.1741E-05	1.5029E-03
GJR-GARCH	2.0097E-05	2.1426E-05	7.3402E-03	3.6346E-05	2.0850E-03
TrGARCH	1.7785E-05	1.8724E-05	3.2888E-03	3.2550E-05	9.4212E-04
APGARCH	2.2457E-05	2.5524E-05	1.0844E-02	7.7428E-05	3.0841E-03
AVGARCH	1.7085E-05	2.1887E-05	2.2334E-03	4.0650E-05	6.4773E-04
FIGARCH	1.9699E-05	2.4777E-05	5.4591E-03	4.3747E-05	1.5580E-03
LSTM-GARCH	2.5959E-05	6.4459E-05	1.7169E-04	3.8912E-05	8.0264E-05
T-GARCH	2.9425E-05	3.9877E-05	1.6371E-04	4.0953E-05	7.1128E-05
TL-GARCH	4.1748E-05	5.8189E-05	1.5675E-04	5.5462E-05	7.9242E-05
MT-GARCH	3.0656E-05	4.3549E-05	1.5675E-04	4.4545E-05	7.5677E-05
MTL-GARCH	5.0336E-05	5.9547E-05	1.4450E-04	6.4434E-05	7.9965E-05

Notes: Same as Table 9.

Table 13
RMSE by model and year for the test set of Reliance.

Model	2018	2019	2020	2021	2018–2021
GARCH	9.4107E-04	4.8983E-04	1.0361E-02	6.7680E-04	3.4966E-03
GJR-GARCH	1.1525E-03	3.2151E-04	1.0373E-02	6.9471E-04	3.4927E-03
TrGARCH	4.7537E-04	3.3466E-04	6.0048E-03	7.4591E-04	2.1187E-03
APGARCH	1.7795E-03	3.6508E-04	1.4334E-02	7.2643E-04	4.7768E-03
AVGARCH	4.3436E-04	3.8449E-04	1.0084E-02	7.2842E-04	3.3125E-03
FIGARCH	9.6922E-04	5.4718E-04	1.4491E-02	7.9872E-04	4.7546E-03
LSTM-GARCH	9.6964E-05	6.8086E-05	3.4430E-04	1.0689E-04	1.6351E-04
T-GARCH	1.2433E-04	6.6591E-05	4.3715E-04	6.7881E-05	1.8496E-04
TL-GARCH	1.1080E-04	6.1806E-05	3.8502E-04	7.2254E-05	1.6718E-04
MT-GARCH	1.0349E-04	6.2104E-05	3.4021E-04	6.7820E-05	1.5181E-04
MTL-GARCH	1.0102E-04	7.9086E-05	3.6300E-04	7.6460E-05	1.6506E-04

Notes: Same as Table 9.

MT-GARCH and MTL-GARCH models that showed lower RMSEs for Euro-USD, FTSE 100, Reliance and Samsung for the entire test period.

The RMSE of the overall test period for S&P 500, FTSE 100, Reliance and Samsung clearly show that the hybrid models outperform the GARCH-type models due to the enhanced forecasting accuracy compared to the individual GARCH-type models. This is consistent with Kristjanpoller et al. (2014), Lu et al. (2016), Kim and Won (2018) and Ramos-Pérez et al. (2021). Furthermore, analyzing the RMSE of all the models for the year 2020 when the COVID-19 pandemic hit, we can observe that the hybrid models perform significantly better than the individual GARCH-type models. This shows that even during periods of high volatility, the performance of the hybrid Neural-network based models is better than the individual GARCH-type models, thereby leading to more accurate stock market volatility forecasts. Similarly in 2018, when the overall volatility was quite low, the hybrid models have consistently shown lower RMSE, thus indicating that such models can be reliable in periods of both high and low volatility.

The Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) are two commonly used metrics to evaluate the accuracy of forecasting models. Both measures are widely regarded as good indicators of forecast accuracy, but they have some fundamental differences.

MAE measures the absolute difference between the forecasted values and the actual values. It is calculated by taking the average of the absolute values of the errors. MAE has the advantage of being easy to interpret, and it treats all errors equally, regardless of their direction. Moreover, MAE is less sensitive to outliers than other metrics like RMSE. On the other hand, MAPE expresses the absolute error as a percentage of the actual value. It is calculated by taking the average of the absolute values of the percentage errors. However, MAPE has a significant disadvantage that it produces infinite or undefined values for zero or close-to-zero actual values, which limits its use in certain scenarios (Kim S. & Kim H., 2016).

For this reason, in our paper, we are using MAE as a measure of

Table 14
RMSE by model and year for the test set of Samsung.

Model	2018	2019	2020	2021	2018–2021
GARCH	6.2048E-04	2.0439E-04	2.5508E-03	3.0351E-04	9.5339E-04
GJR-GARCH	5.8938E-04	1.9815E-04	9.1365E-04	3.2506E-04	5.0186E-04
TrGARCH	5.2844E-04	2.5384E-04	7.2422E-04	3.4148E-04	4.5789E-04
APGARCH	5.1584E-04	2.1767E-04	9.6687E-04	3.2267E-04	5.0731E-04
AVGARCH	5.0408E-04	2.5760E-04	1.7508E-03	2.9410E-04	7.2379E-04
FIGARCH	3.6940E-04	2.4649E-04	2.1292E-03	3.2949E-04	8.0857E-04
LSTM-GARCH	6.8382E-05	1.6374E-04	1.5488E-04	7.1641E-05	1.1908E-04
T-GARCH	5.9088E-05	5.8309E-05	1.6658E-04	5.8369E-05	8.8208E-05
TL-GARCH	7.4874E-05	4.1083E-05	1.7347E-04	6.3480E-05	8.9708E-05
MT-GARCH	5.7877E-05	9.5280E-05	1.3417E-04	5.8181E-05	8.9142E-05
MTL-GARCH	5.6152E-05	5.6457E-05	1.7806E-04	5.7205E-05	8.9992E-05

Notes: Same as Table 9.

forecast accuracy (see Tables 15–20). While MAPE has its advantages, the risk of producing undefined values for small actual values makes it less practical for some applications. In contrast, MAE provides a reliable and interpretable measure of forecast accuracy, making it a suitable choice for our analysis.

Table 15
MAE by model and year for the test set of EURO-USD.

Model	2018	2019	2020	2021	2018–2021
GARCH	2.2988E-03	1.6729E-03	2.3155E-03	2.0790E-03	2.0735E-03
GJR-GARCH	2.2973E-03	1.6981E-03	2.2167E-03	2.1321E-03	2.0668E-03
TrGARCH	2.2956E-03	1.6528E-03	2.1398E-03	2.1223E-03	2.0299E-03
APGARCH	2.2994E-03	1.6633E-03	2.2571E-03	2.0911E-03	2.0579E-03
AVGARCH	2.2837E-03	1.6337E-03	2.2292E-03	2.0212E-03	2.0206E-03
FIGARCH	2.2906E-03	1.6425E-03	2.1884E-03	2.0367E-03	2.0173E-03
LSTM-GARCH	1.0005E-02	7.4792E-03	3.9078E-03	6.9811E-03	6.7898E-03
T-GARCH	6.6619E-03	6.4146E-03	5.1694E-03	5.6276E-03	5.8974E-03
TL-GARCH	3.3160E-03	4.6099E-03	3.9381E-03	4.3127E-03	4.1059E-03
MT-GARCH	8.0298E-03	7.9711E-03	4.9889E-03	5.4686E-03	6.4684E-03
MTL-GARCH	3.3563E-03	3.9476E-03	3.7252E-03	3.7154E-03	3.7240E-03

Notes:
a MAE (Mean Absolute Error) for all the hybrid Neural-network models as well as the individual traditional GARCH-type models, for each year in the test set and for the overall test period.
b The GARCH model is the GARCH (1,1) model; GJR-GARCH model is Glosten-Jagannathan-Runkle GARCH model; APGARCH model is the Asymmetric Power GARCH (1,1) model; TrGARCH model is the Threshold GARCH (1,1) model, FIGARCH model is fractionally integrated GARCH model and AVGARCH model is absolute value GARCH model.
c LSTM-GARCH model is The Long short-term Memory-Generalized Auto Regressive Conditional Heteroskedasticity model; T-GARCH model is the Transformer-GARCH model, TL-GARCH model is the Transformer-LSTM-GARCH model, MT-GARCH model is the Multi-Transformer-GARCH model and the MTL-GARCH model is the Multi-Transformer-LSTM-GARCH model.

Table 16
MAE by model and year for the test set of AUD-USD.

Model	2018	2019	2020	2021	2018–2021
GARCH	2.5943E-03	1.7980E-03	3.1572E-03	2.0613E-03	2.3920E-03
GJR-GARCH	2.6419E-03	1.8110E-03	3.3094E-03	2.1168E-03	2.4628E-03
TrGARCH	2.7914E-03	1.6682E-03	3.0848E-03	2.1040E-03	2.3842E-03
APGARCH	2.8206E-03	1.7934E-03	3.5136E-03	2.1468E-03	2.5541E-03
AVGARCH	2.7782E-03	1.6920E-03	2.8811E-03	1.9940E-03	2.3004E-03
FIGARCH	2.7457E-03	1.7789E-03	2.9040E-03	2.1380E-03	2.3638E-03
LSTM-GARCH	1.0048E-02	1.1137E-02	4.2944E-03	6.5996E-03	7.8051E-03
T-GARCH	3.6678E-03	4.0815E-03	3.6430E-03	3.3302E-03	3.6813E-03
TL-GARCH	3.6119E-03	3.7807E-03	3.8410E-03	3.2090E-03	3.6099E-03
MT-GARCH	5.6509E-03	6.0716E-03	3.8312E-03	3.5592E-03	4.6958E-03
MTL-GARCH	3.8110E-03	4.4709E-03	3.9857E-03	4.4810E-03	4.2123E-03

Notes: Same as Table 15.

Table 17
MAE by model and year for the test set of Reliance.

Model	2018	2019	2020	2021	2018–2021
GARCH	2.1402E-02	1.7969E-02	5.5488E-02	2.1722E-02	3.0406E-02
GJR-GARCH	2.1132E-02	1.5653E-02	5.2741E-02	2.2235E-02	2.9044E-02
TrGARCH	1.5036E-02	1.6194E-02	4.3277E-02	2.2634E-02	2.5498E-02
APGARCH	2.6007E-02	1.6129E-02	5.9724E-02	2.3779E-02	3.2433E-02
AVGARCH	1.6199E-02	1.7286E-02	5.3609E-02	2.2377E-02	2.8940E-02
FIGARCH	2.2733E-02	1.9027E-02	6.8916E-02	2.2910E-02	3.5148E-02
LSTM-GARCH	7.2510E-03	5.8794E-03	1.2033E-02	7.9341E-03	8.4101E-03
T-GARCH	8.5254E-03	6.0224E-03	1.2345E-02	6.1033E-03	8.3043E-03
TL-GARCH	7.7119E-03	5.6973E-03	1.1681E-02	6.7539E-03	8.0468E-03
MT-GARCH	7.5270E-03	5.5565E-03	1.1722E-02	6.5723E-03	7.9377E-03
MTL-GARCH	7.5231E-03	5.5363E-03	1.1704E-02	6.9016E-03	8.0105E-03

Notes: Same as Table 15.

The MAE of the overall test period for all equity based as S&P 500, FTSE 100, Reliance and Samsung clearly show that the hybrid models outperform the GARCH-type models due to the enhanced forecasting accuracy compared to the individual GARCH-type models. This is consistent with RMSE results and the literature studied. Furthermore, analyzing the MAE for the year 2020 when the COVID-19 pandemic hit, we can observe that the hybrid models perform significantly better than the individual GARCH-type models, again similar to the results shown by RMSE, confirming that the hybrid models can show consistent results even during the periods of high volatility. Again, for equity-based assets, during the relatively less volatile period in 2018, the RMSE for hybrid transformer-based models has been consistently lower, confirming these hybrid models can be relied upon in less volatile conditions as well.

However, for Euro-USD and AUD-USD, the GARCH-type models perform slightly better, highlighting one of the few drawbacks of deep learning based models which are prone to overfitting on the training data set. However, compared to RMSE results, the MAE for the hybrid

Table 18
MAE by model and year for the test set of Samsung.

Model	2018	2019	2020	2021	2018–2021
GARCH	2.3667E-02	1.3075E-02	2.6177E-02	1.3651E-02	1.8825E-02
GJR-GARCH	2.2995E-02	1.2963E-02	1.9062E-02	1.4150E-02	1.6832E-02
TrGARCH	2.1800E-02	1.4813E-02	1.7938E-02	1.4321E-02	1.6849E-02
APGARCH	2.1048E-02	1.3694E-02	1.9340E-02	1.4330E-02	1.6787E-02
AVGARCH	2.1253E-02	1.5064E-02	2.5173E-02	1.3578E-02	1.8614E-02
FIGARCH	1.8030E-02	1.4696E-02	2.7220E-02	1.3812E-02	1.8528E-02
LSTM-GARCH	6.4830E-03	8.0132E-03	8.0489E-03	6.6768E-03	7.3903E-03
T-GARCH	5.8069E-03	5.9359E-03	8.3492E-03	6.2785E-03	6.6543E-03
TL-GARCH	6.1661E-03	5.0475E-03	8.5634E-03	6.5759E-03	6.6244E-03
MT-GARCH	5.8537E-03	6.0878E-03	7.6417E-03	6.1596E-03	6.4892E-03
MTL-GARCH	5.8562E-03	5.6806E-03	8.5849E-03	6.1524E-03	6.6320E-03

Notes: Same as Table 15.

Table 19
MAE by model and year for the test set of S&P 500.

Model	2018	2019	2020	2021	2018–2021
GARCH	5.3522E-03	4.3990E-03	3.7747E-02	4.4557E-03	1.3890E-02
GJR-GARCH	6.7732E-03	4.1259E-03	4.3187E-02	5.0091E-03	1.5707E-02
TrGARCH	6.5067E-03	3.8317E-03	3.9766E-02	4.9634E-03	1.4610E-02
APGARCH	7.2904E-03	4.4645E-03	4.5119E-02	5.1983E-03	1.6480E-02
AVGARCH	4.3793E-03	3.7517E-03	3.1045E-02	4.6808E-03	1.1730E-02
FIGARCH	5.0516E-03	4.3306E-03	3.7510E-02	4.4486E-03	1.3753E-02
LSTM-GARCH	6.2614E-03	5.3871E-03	8.9030E-03	5.9679E-03	6.6659E-03
T-GARCH	5.9843E-03	5.2975E-03	1.0097E-02	5.1881E-03	6.7209E-03
TL-GARCH	5.5664E-03	5.0553E-03	1.0424E-02	4.9177E-03	6.6019E-03
MT-GARCH	5.1506E-03	4.9734E-03	1.0318E-02	5.2626E-03	6.5730E-03
MTL-GARCH	6.3544E-03	4.7956E-03	9.6066E-03	5.5864E-03	6.6029E-03

Notes: Same as Table 15.

Table 20
MAE by model and year for the test set of FTSE.

Model	2018	2019	2020	2021	2018–2021
GARCH	3.3914E-03	3.3460E-03	2.7058E-02	5.2098E-03	1.0506E-02
GJR-GARCH	3.6138E-03	3.0100E-03	2.7843E-02	4.9639E-03	1.0605E-02
TrGARCH	3.3951E-03	2.9304E-03	2.0912E-02	4.6926E-03	8.5290E-03
APGARCH	3.7810E-03	3.1386E-03	3.2026E-02	6.5203E-03	1.2264E-02
AVGARCH	3.2903E-03	3.1940E-03	1.9506E-02	5.0362E-03	8.2823E-03
FIGARCH	3.6006E-03	3.1941E-03	2.8548E-02	5.4124E-03	1.0973E-02
LSTM-GARCH	4.0214E-03	4.7979E-03	8.7359E-03	4.7402E-03	5.7515E-03
T-GARCH	4.1951E-03	4.2840E-03	7.9974E-03	5.2794E-03	5.5790E-03
TL-GARCH	4.6559E-03	5.3844E-03	7.9170E-03	6.1266E-03	6.1688E-03
MT-GARCH	4.3248E-03	4.1913E-03	8.2701E-03	5.2481E-03	5.6439E-03
MTL-GARCH	4.8865E-03	5.2434E-03	7.7301E-03	6.0565E-03	6.0986E-03

Notes: Same as Table 15.

methods are consistently lower or comparable to the GARCH-type models, thus suggesting that such hybrid models are very strong alternatives to the traditional GARCH based models.

Expected shortfall (ES), which is another measure of risk that is recommended by the Basel Committee, is found to be a coherent measure of risk, especially due to its sub-additive nature i.e., the risk of a portfolio would never be greater than the sum of the standalone risk components that is measured using VaR (Acerbi, Nordio & Sirtori, 2001; Acerbi and Tasche, 2002; Wang and Zitikis, 2021; Wei, 2021). ES is the average amount that may be lost over a certain period of time if the loss is greater than a certain percentile of the loss distribution. For instance, if a 95 percent confidence interval for ten days is considered, the ES would be the average amount that may be lost over a ten-day period, assuming that the loss is greater than 95th percentile of the loss distribution.

Fig. 5 shows the expected shortfall for all the six financial assets during the test period. We observe that the MTL-GARCH model is the hybrid neural-network model that provides the most conservative estimate of expected shortfall for all the six financial assets throughout the test period. During the highly volatile COVID-19 period, we can observe models like MTL-GARCH show heavy spikes in the ES graphs, which suggests that MTL-GARCH is one of the only models to capture the Expected Shortfall accurately. In fact, it is the lower variance in error due to the Multi-Transformer architecture helped to predict the ES more accurately than the other hybrid neural-network models.

4.3. Value-at-risk (VaR) Backtesting- Christoffersen test

Market risk is risk of losses in positions resulting from fluctuations in market prices. VaR is one of the most important gauges of financial risk. VaR is an estimate of the amount of value a portfolio may lose over a certain time period with a specified degree of confidence. For instance, if a portfolio's 95 percent VaR for one day is $10 million, there is a 95 percent probability that the portfolio would lose less than $10 million the next day. In other words, only 5% of the time (or once every 20 days) do portfolio losses reach $10,000,000. Consequently, VaR (value-at-risk) is an estimate of the amount of value a portfolio may lose over a certain time period with a specified degree of confidence. VaR backtesting tools assess the accuracy of VaR models. The Financial Crisis of 2007–2008 and the COVID-19 outbreak resulted in substantial losses for banks and insurance companies. In addition, they emphasize the significance of using correct equity risk models and a risk management function capable of implementing successful hedging methods. Forecasts of market volatility play a crucial part in the estimate of equity risk and, therefore, in the management decisions made by financial organizations.

Christoffersen (1998) proposed a test to measure whether the probability of observing an exception on a particular day depends on whether an exception occurred on the previous day. The Christoffersen test facilitates concurrent examination of coverage and independence hypotheses, providing a comprehensive assessment of VaR model performance. This approach allows for individual testing of each hypothesis, aiding in the identification of specific areas of model failure. Unlike the unconditional probability of observing an exception, Christoffersen's test measures the dependency between consecutive days only. Tables 21–26 show the Christofferson test values (p-values) for each volatility model and for each year in the testing set, for all the six assets considered. The Christofferson test focuses on not only the total

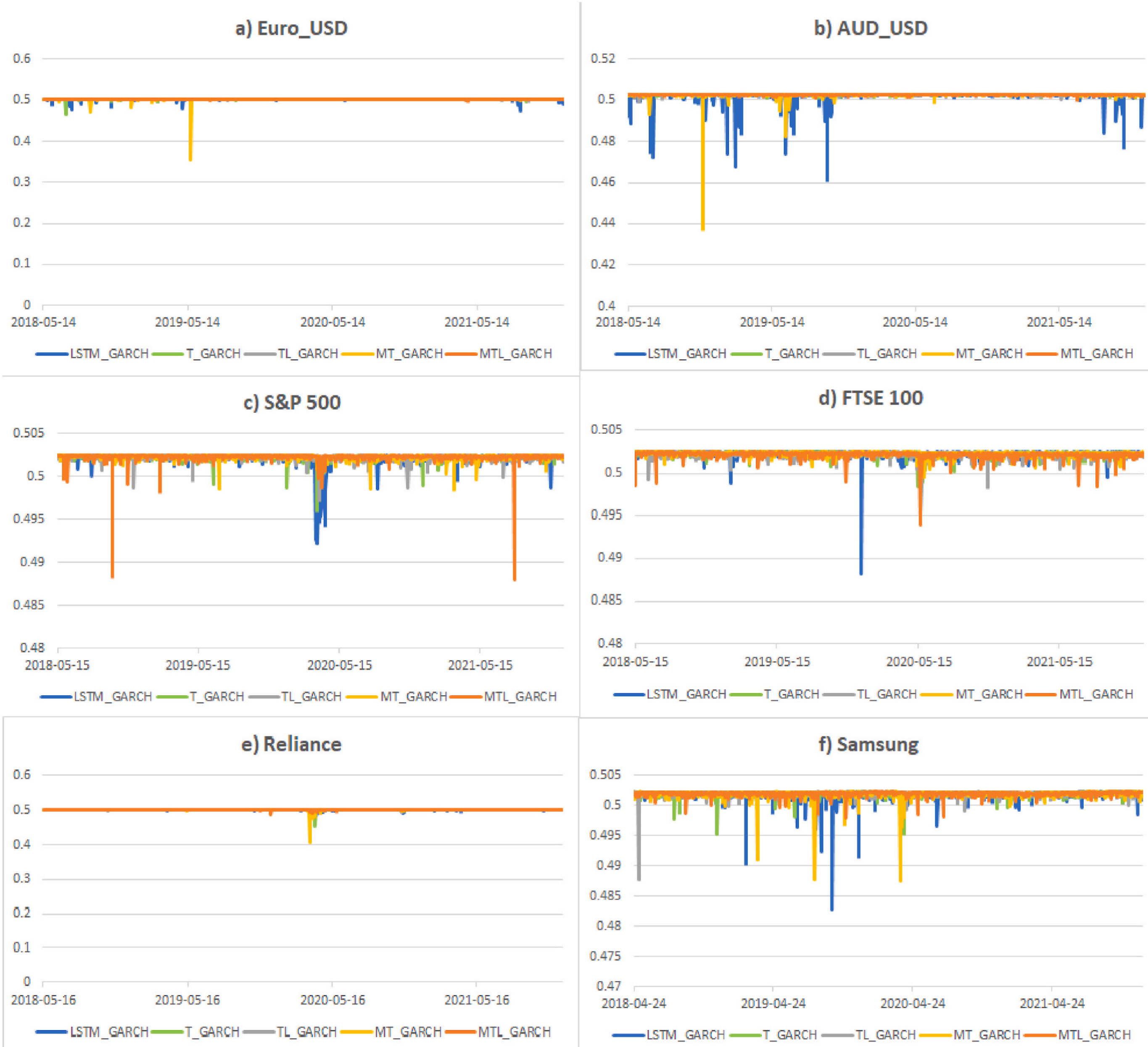

Fig. 5. Expected shortfall for Euro-USD, AUD-USD, S&P 500 Index, FTSE 100 Index, Reliance Industries Ltd. and Samsung Electronics Co Ltd.

number of VaR exceedances but also on the number of consecutive VaR exceedances. We select a risk measure and confidence level of 99%, which is in line with what the regulators prescribe and use (Samanta et al., 2010). We observe from Tables 21–26 that in the 2018–2021 column which shows the Christofferson test values for the entire test period, all the traditional GARCH-type models and three to four of the five hybrid neural network models show appropriate risk measures, since their p-values are greater than 0.01. This indicates that the results shown by these models during the test period can be considered appropriate. Even during the year 2020 when the uncertainty due to the COVID-19 pandemic was at its peak, the risk measures produced by these models passed the Christofferson test. This shows that both the Transformer-based models and the GARCH-type models which are being given as an input to these models are producing results that are appropriate, even during periods of high uncertainty.

Thus, it is inferred that the hybrid models based on Multi-Transformer and Transformer layers are more accurate and, hence, they lead to more appropriate risk measures than other autoregressive algorithms or hybrid models based on feed forward layers or long short-term memory cells.

5. Conclusions

Transformers have a multi-head attention-based mechanism which has achieved state-of-the-art-results for various tasks in Natural Language Processing (NLP) (Vaswani et al., 2017; Vig & Belenkov, 2019). To be able to use these transformer layers with time-series data, there are two main components that have been added in this study which include positional encoding and the multi-head attention mechanism (Ramos-Pérez et al., 2021). The Multi-Transformer architecture is a variant of the Transformer architecture which takes T random samples of the input data and has T heads of multi-attention, one corresponding to each random sample of input data. In addition, the bagging mechanism is applied only to the attention mechanism part of the

Table 21
Christofferson test values by model and year for EURO-USD.

Model	2018	2019	2020	2021	2018–2021
GARCH	5.7571E-01	2.2249E-02	2.1994E-02	6.8206E-01	4.5001E-02
GJR-GARCH	5.7571E-01	2.2249E-02	2.5077E-01	3.4299E-01	2.2146E-01
TrGARCH	5.7571E-01	2.5301E-01	2.5077E-01	3.4299E-01	3.7745E-01
APGARCH	2.2146E-01	2.2146E-01	2.2146E-01	2.2146E-01	2.2146E-01
AVGARCH	5.7571E-01	2.2249E-02	2.1994E-02	6.8206E-01	4.5001E-02
FIGARCH	5.7571E-01	2.2249E-02	2.1994E-02	6.8206E-01	4.5001E-02
LSTM-GARCH	6.8583E-02	2.2249E-02	2.5077E-01	3.4299E-01	1.1014E-01
T-GARCH	6.8583E-02	2.2249E-02	2.1994E-02	2.4982E-02	1.3960E-05
TL-GARCH	5.7571E-01	2.2249E-02	2.5077E-01	2.7644E-01	1.4331E-02
MT-GARCH	6.8583E-02	2.2249E-02	2.5077E-01	2.7644E-01	3.2507E-03
MTL-GARCH	7.2977E-01	2.5301E-01	2.1994E-02	2.7644E-01	4.5001E-02

Notes:
a The Christofferson test values (p-values) for all the hybrid Neural-network models as well as the individual traditional GARCH-type models, for each year in the test set and for the overall test period.
b The GARCH model is the GARCH (1,1) model; GJR-GARCH model is Glosten-Jagannathan-Runkle GARCH model; APGARCH model is the Asymmetric Power GARCH (1,1) model; TrGARCH model is the Threshold GARCH (1,1) model, FIGARCH model is fractionally integrated GARCH model and AVGARCH model is absolute value GARCH model.
c LSTM-GARCH model is The Long short-term Memory-Generalized Auto Regressive Conditional Heteroskedasticity model; T-GARCH model is the Transformer-GARCH model, TL-GARCH model is the Transformer-LSTM-GARCH model, MT-GARCH model is the Multi-Transformer-GARCH model and the MTL-GARCH model is the Multi-Transformer-LSTM-GARCH model.

Table 22
Christofferson test values by model and year for AUD-USD.

Model	2018	2019	2020	2021	2018–2021
GARCH	5.7571E-01	2.5301E-01	1.6384E-01	2.7644E-01	5.5057E-01
GJR-GARCH	5.7571E-01	2.5301E-01	6.0052E-02	2.7644E-01	6.6226E-01
TrGARCH	5.7571E-01	2.5301E-01	3.8097E-01	2.7644E-01	3.7745E-01
APGARCH	5.5057E-01	5.5057E-01	5.5057E-01	5.5057E-01	5.5057E-01
AVGARCH	5.7571E-01	2.5301E-01	3.8097E-01	2.4982E-02	2.2146E-01
FIGARCH	5.7571E-01	2.5301E-01	2.7792E-02	2.7644E-01	2.5994E-02
LSTM-GARCH	7.2977E-01	2.5301E-01	6.0052E-02	2.7644E-01	6.1393E-01
T-GARCH	5.7571E-01	2.2249E-02	1.6382E-02	6.8206E-01	8.8644E-02
TL-GARCH	8.8463E-03	2.2249E-02	1.6382E-02	3.4299E-01	4.0221E-03
MT-GARCH	7.2977E-01	2.5301E-01	6.6921E-04	6.8206E-01	5.0405E-03
MTL-GARCH	6.8583E-02	2.2249E-02	2.5565E-02	2.7644E-01	1.1762E-02

Notes: Same as Table 21.

Table 23
Christofferson test values by model and year for S&P 500.

Model	2018	2019	2020	2021	2018–2021
GARCH	6.5668E-01	7.1146E-01	3.0288E-01	7.9229E-01	6.5283E-01
GJR-GARCH	6.4390E-01	7.1146E-01	6.3353E-01	2.3199E-03	5.1045E-01
TrGARCH	6.4390E-01	7.1146E-01	6.3353E-01	7.9229E-01	6.5283E-01
APGARCH	5.1045E-01	5.1045E-01	5.1045E-01	5.1045E-01	5.1045E-01
AVGARCH	6.5668E-01	7.1146E-01	3.0288E-01	7.9229E-01	6.5283E-01
FIGARCH	6.4390E-01	7.1146E-01	3.0288E-01	5.8657E-01	3.5056E-01
LSTM-GARCH	6.5727E-03	2.7889E-01	1.3318E-04	7.9229E-01	3.5092E-03
T-GARCH	5.6947E-06	2.7889E-01	1.9432E-03	3.2673E-02	6.9175E-05
TL-GARCH	3.8808E-03	7.1146E-01	2.0369E-07	7.9229E-01	1.5460E-05
MT-GARCH	2.5098E-01	6.7816E-01	2.2798E-06	3.3841E-01	3.6765E-04
MTL-GARCH	3.0299E-04	7.1146E-01	1.2432E-06	3.3841E-01	1.3067E-05

Notes: Same as Table 21.

Table 24
Christofferson test values by model and year for FTSE 100.

Model	2018	2019	2020	2021	2018–2021
GARCH	2.6263E-01	7.9554E-01	1.0555E-01	5.6472E-01	1.0220E-01
GJR-GARCH	8.1335E-02	5.8222E-01	1.0555E-01	5.6472E-01	2.6704E-02
TrGARCH	8.1335E-02	7.9554E-01	1.0555E-01	5.6472E-01	5.3921E-02
APGARCH	1.0220E-01	1.0220E-01	1.0220E-01	1.0220E-01	1.0220E-01
AVGARCH	2.6263E-01	1.9498E-02	3.4131E-02	5.6472E-01	1.8043E-02
FIGARCH	6.6075E-01	7.9554E-01	1.0555E-01	8.0785E-01	2.9875E-01
LSTM-GARCH	2.0503E-02	7.9554E-01	3.6011E-07	8.0785E-01	2.3026E-05
T-GARCH	6.6075E-01	3.4138E-01	1.1611E-08	3.5352E-01	2.5274E-04
TL-GARCH	8.0474E-02	3.4138E-01	2.5221E-05	3.5352E-01	4.6677E-02
MT-GARCH	2.6263E-01	9.9055E-02	1.1611E-08	5.6472E-01	4.2525E-07
MTL-GARCH	8.1335E-02	3.4138E-01	1.1567E-04	3.4650E-02	1.8043E-02

Notes: Same as Table 21.

Multi-Transformer architecture.

This study introduces a more accurate model for the prediction of volatility in assets like the foreign exchange rate, the stock market index and the stock price of companies by using hybrid Neural-network models that combine Transformer and Multi-Transformer layers with the GARCH model and the LSTM model. For this purpose, six assets were considered, namely, Euro-US Dollar FX Spot Rate (Euro-USD), Australian Dollar-US Dollar FX Spot Rate (AUD-USD), S&P 500 Index, FTSE 100 Index, Reliance Industries Ltd. and Samsung Electronics Co Ltd. for the period January 2005 to December 2021. Out of all the models considered, both the individual traditional GARCH-type models and the hybrid Neural-network models, the latter was found to significantly outperform the former, especially the Transformer-based models (Kristjanpoller et al., 2014; Lu et al., 2016; Kim and Won, 2018; Ramos-Pérez et al., 2021), as they showed considerably lower RMSE for the test period spanning from January 2017 to December 2021. In addition, the Transformer-based models also showed significantly lower RMSE compared to the traditional GARCH-type models, both during low volatile conditions and even during the highly volatile environments like the COVID-19 pandemic. Furthermore, within the hybrid Neural-network models, the MT-GARCH and MTL-GARCH model showed lower RMSEs for the entire test period for Euro-USD, FTSE 100,

Table 25
Christofferson test values by model and year for Reliance.

Model	2018	2019	2020	2021	2018–2021
GARCH	6.2680E-01	2.9147E-01	6.8593E-01	3.6321E-02	3.2115E-01
GJR-GARCH	6.2680E-01	2.9147E-01	6.8593E-01	3.6321E-02	3.2115E-01
TrGARCH	6.2680E-01	2.6786E-02	6.8593E-01	3.6321E-02	1.7180E-01
APGARCH	3.2115E-01	3.2115E-01	3.2115E-01	3.2115E-01	3.2115E-01
AVGARCH	1.7599E-02	2.9147E-01	3.4640E-01	3.6321E-02	5.7034E-02
FIGARCH	1.7599E-02	2.9147E-01	1.4352E-01	3.6606E-01	8.7683E-02
LSTM-GARCH	1.7599E-02	2.6786E-02	3.4640E-01	3.6606E-01	5.7034E-02
T-GARCH	5.1103E-04	6.5823E-01	2.5809E-03	2.3998E-01	3.5813E-05
TL-GARCH	1.3385E-02	2.9147E-01	1.5470E-02	3.6606E-01	5.4421E-02
MT-GARCH	6.1812E-03	7.3072E-01	1.4352E-01	8.1881E-01	3.6807E-02
MTL-GARCH	1.3385E-02	2.9147E-01	5.0618E-02	8.1881E-01	5.4421E-02

Notes: Same as Table 21.

Table 26
Christofferson test values by model and year for Samsung.

Model	2018	2019	2020	2021	2018–2021
GARCH	5.7848E-02	2.5301E-01	3.8097E-01	2.4982E-02	1.0134E-01
GJR-GARCH	5.7848E-02	2.5301E-01	7.2269E-01	2.7644E-01	1.0134E-01
TrGARCH	5.7848E-02	2.2249E-02	7.2269E-01	2.7644E-01	4.0883E-02
APGARCH	1.2854E-02	1.2854E-02	1.2854E-02	1.2854E-02	1.2854E-02
AVGARCH	5.7848E-02	2.2249E-02	7.2269E-01	2.4982E-02	1.2854E-02
FIGARCH	5.7848E-02	2.2249E-02	7.2269E-01	2.4982E-02	1.2854E-02
LSTM-GARCH	5.7848E-02	2.2249E-02	6.6499E-01	2.4982E-02	2.8781E-03
T-GARCH	5.7848E-02	2.2249E-02	8.0053E-03	2.7644E-01	4.4760E-02
TL-GARCH	5.7848E-02	2.5301E-01	8.0053E-03	2.7644E-01	6.8186E-02
MT-GARCH	5.1183E-01	2.2249E-02	7.2269E-01	2.4982E-02	4.0883E-02
MTL-GARCH	5.7848E-02	6.6885E-01	3.1636E-03	2.4982E-02	8.8048E-02

Notes: Same as Table 21.

Reliance and Samsung. The same can be observed while using MAE metric as well, with MT-GARCH and MTL-GARCH showing lower MAEs. This shows that the mechanism of bagging, which was applied to the attention mechanism part of the Multi-Transformer architecture, reduced the variance in the noisy data of the daily returns of these assets, thereby decreasing the RMSE and improving the accuracy of the volatility forecasts. The results of the Christoffersen's test help in confirming that the results shown by these models during the test period can be considered appropriate. Moreover, the MTL-GARCH model also showed the most conservative estimate of expected shortfall.

Therefore, this study provides three primary contributions. First, Multi-Transformer layers are modified to predict the volatility of financial markets. Second, this research indicates that the combination of Transformer and Multi-Transformer layers with other models results in more precise volatility forecasting models. Thirdly, the suggested stock volatility models produce suitable risk assessments during periods of low and high volatility.

This study's findings may be of particular relevance to risk management professionals who can manage the risk component in their stock market investment decisions in a better way, using the improved stock market volatility forecasts made by the hybrid Multi-Transformer architectures. In addition, these findings can also be relevant to policymakers in order to more accurately decide the limits on investments and other related parameters as well as modify the policies corresponding to them whenever required, based on the volatility forecasts of the hybrid Multi-Transformer models, which are more accurate even during highly volatile environments compared to the traditional GARCH-type models.

Future studies can extend this analysis by combining the hybrid Transformer and Multi-Transformer models with machine learning algorithms such as decision trees and random forest. In addition, the key assumptions of the attention mechanism in the Transformer and Multi-Transformer architectures can also be modified to produce more accurate market volatility forecasts. Furthermore, since volatilities play an important role in the valuation of derivatives (Black and Scholes, 1973; Figlewski, 1997), therefore, the application of the hybrid Transformer and Multi-Transformer models for derivative valuation purposes can also be explored.

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Availability of data and material (data transparency)

The data will be available upon request.

Code availability (software application or custom code)

The code will be available upon reasonable request.

Ethics statement

The study did not require ethical approval.

CRediT authorship contribution statement

Aswini Kumar Mishra: Supervision, Conceptualization, Validation, Writing – review & editing. **Jayashree Renganathan:** Methodology, Data curation, Software, Writing – original draft. **Aaryaman Gupta:** Data curation, Software, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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